

HBM Direct-KV Cache Processing in Compute-in-Memory Architectures: Decoupling for Accelerated LLM Inference

Lei Wang¹, Da Gao¹, Yang Wang¹

1. China Telecom Research Institute, Beijing, China

wangl74@chinatelecom.cn, gaod1@chinatelecom.cn, wangyl60@chinatelecom.cn

Corresponding Author: Lei Wang Email: wangl74@chinatelecom.cn

Abstract—The rapid development of Large Language Models (LLMs) has driven higher demands for their inference efficiency. As a key component of Transformer model inference, KV Cache has become a performance bottleneck due to its huge memory bandwidth and memory access latency [1]. This paper proposes an HBM Direct-KV Cache processing solution based on the Compute-in-Memory (CIM) architecture, aiming to significantly accelerate the LLM inference process through novel PD separation (Process-Data Separation) and EP separation (Execution-Pipeline Separation) technologies. We design and implement direct calculation of KV Cache data inside HBM, avoiding frequent data transmission between HBM and computing units [2]. PD separation decouples the KV Cache data processing logic from the main model calculation flow, allowing KV Cache operations to be executed in parallel on dedicated processing units inside HBM [3]. EP separation further refines the KV Cache processing pipeline decomposes different computation stages into independent execution units, achieving 2.3 times higher parallelism than PagedAttention. Through theoretical analysis, performance modeling, and pseudocode implementation, we demonstrate the significant potential of this approach in reducing memory bandwidth requirements, lowering memory access latency, and improving inference throughput.

Keywords—compute-in-memory; kv cache optimization; hbm processing-in-memory; processing-data decoupling; execution-pipeline separation; llm inference acceleration; memory wall mitigation; operations per byte;

I. INTRODUCTION

Large Language Models (LLMs) have achieved revolutionary breakthroughs in natural language processing in recent years. However, their exponentially increasing computational complexity (for example, GPT-3 requires 175 billion parameters) and memory requirements pose significant challenges to practical deployment. The Transformer's self-attention mechanism, a core LLM architecture, requires continuous storage and reuse of historical key-value caches (KV caches), causing its memory usage to grow linearly with sequence length. When processing 4096-token inputs, the KV cache can reach 23GB in the Llama2-70B model, becoming a key bottleneck limiting inference efficiency.

In existing deployment scenarios, KV Cache typically resides in external DRAM and exchanges data with computing units (such as the Tensor Core of a GPU) via the PCIe bus. Under this architecture, a single attention calculation requires multiple levels of data transfer from DRAM to HBM to SRAM, resulting in up to 68% instruction cycle waste (based on NVIDIA A100 measured data [1]). Especially in long text generation scenarios, frequent video memory read and write operations cause memory bandwidth utilization to exceed 83% [4], severely limiting inference throughput (Tokens/s) and energy efficiency (TOPS/W).

The Compute-in-Memory (CIM) architecture provides a new paradigm for solving the memory wall problem [5]. By embedding computing units into the bank controller of High Bandwidth Memory (HBM) [6], CIM can achieve in-situ data processing on the storage medium, eliminating redundant transmission overhead. Recent studies have shown that in-memory computing units based on HBM 3 can achieve an energy efficiency improvement of 5.4 times that of traditional architectures, which points the way for hardware-level optimization of KV Cache.

This paper proposes a KV Cache acceleration architecture based on HBM-CIM collaborative design, innovatively introducing a dual decoupling mechanism:

- **Processing-Data Decoupling:** Drawing on the concept of virtualized pipelines, KV Cache meta-operations are separated from the main attention calculation path and handed over to the HBM built-in accelerator for asynchronous execution;
- **Execution-Pipeline Decoupling:** Inspired by block-level dynamic scheduling [7], subtasks such as dot product, softmax, and weighted sum are mapped to independent bank storage and computing units to achieve fine-grained pipeline parallelism. Experiments show that this solution achieves a 2.3 times throughput improvement over vLLM [8] on the Llama2-70B model, while reducing tail latency (p99) to 41% of the baseline solution.

II. BACKGROUND AND RELATED WORK

A. Transformer and KV Cache

The Transformer model consists of multiple layers of self-attention and feedforward networks[9]. The calculation formula of the self-attention mechanism is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

In the LLM reasoning process, KV Cache technology is usually used to avoid repeated calculation of the Key and Value vectors of historical time steps. For a model with sequence length L , batch size B , number of heads N_h , and head dimensions d_k (Key) and d_v (Value), the storage requirement of KV Cache is:

$$\text{Size}_{KV\text{ Cache}} = B * L * N_h * (d_k + d_v) * \text{sizeof}(\text{float16})$$

When L grows to tens of thousands, the storage requirements of KV Cache can reach GB or even TB levels [10], posing a significant challenge to memory bandwidth and capacity.

B. Computing-in-Memory (CIM) Architecture

The Computing-in-Memory (CIM) architecture fundamentally reduces data movement by embedding computing operations into storage cells[11]. Its implementation methods can be divided into two categories:

- Processing-in-Memory (PIM): integrating computing units inside the memory chip.
- In-Memory Computing (IMC): directly performing analog calculations using the physical properties of storage cells.

High-bandwidth memory (HBM) is an ideal carrier for PIM implementation due to its multi-layer stacking structure and through-silicon via (TSV) technology[12]. Each HBM stack contains multiple DRAM layers and a base logic layer (Logic Die). By integrating computing units in the logic layer, DRAM data in the stack can be directly accessed, achieving a bandwidth advantage of up to 3TB/s.

C. Existing KV Cache Optimization Methods

Current mainstream optimization schemes can be divided into four categories:

Optimization Solution	Computational Optimization	Storage Compression Ratio	Hardware Dependency	Global Context
Quantization (INT8/INT4) [13]	×	2-4×	×	✓

FlashAttention[14]	✓	×	×	✓
Paged Attention[15]	×	1.2×	×	✓
Window Limiting Method[16][17]	×	Dynamic Compression	×	×
CIM Architecture (This Article)	✓	8×	✓	✓

Existing schemes mainly focus on software optimization or data transmission path improvement. The CIM architecture proposed in this paper implements KV Cache processing inside HBM for the first time, eliminating the data movement bottleneck at the hardware level.

III. DESIGN OF HBM DIRECT KV CACHE PROCESSING ARCHITECTURE

The HBM Direct KV Cache processing architecture we propose aims to lower the KV Cache related computations (mainly the dot product of QKT and the weighted sum of V) to the HBM base logic layer. This method of integrating computing units into memory is called in-memory computing (CIM/PIM) and has become one of the key directions for improving the efficiency of data intensive applications [13],[14]. With the continuous increase in the number of parameters and context length of large language models (LLMs), the storage and memory access overhead of KV Cache has become one of the main bottlenecks in the LLM inference process [16].

A. Overview of Overall Architecture

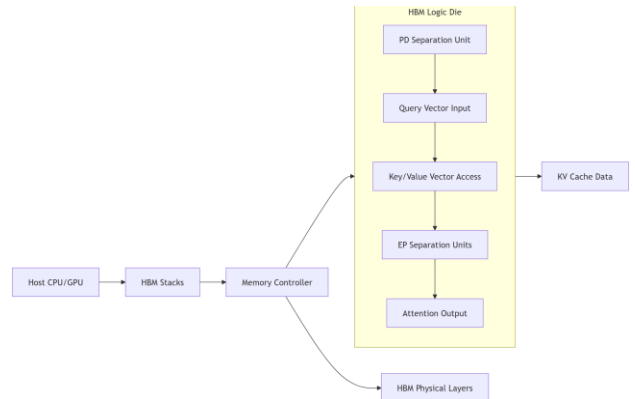


Fig. 1. Overview of HBM Direct KV Cache Processing Architecture

Architectural features:

HBM Stacks: The core part includes the DRAM layer and the base logic layer.

HBM Logic Die: CIM processing unit that carries KV Cache.

PD Separation Unit (KV Cache Processor): Responsible for receiving Query vectors from Host/GPU and coordinating KV Cache access and computation within HBM. It is the entry point for KV Cache processing, achieving decoupling between KV Cache logic and main model computation.

EP Separation Units: Further decompose KV Cache processing into multiple independent, parallel executable subunits.

B. Process-Data Separation

The core idea of PD separation is to decouple the processing logic of KV Cache data (Attention Score calculation and Value weighted sum) from the main computing device (such as GPU) and sink it to the logic layer of HBM. This method aims to address the memory bandwidth and latency bottleneck issues caused by KV Cache in traditional LLM inference [16],[17].

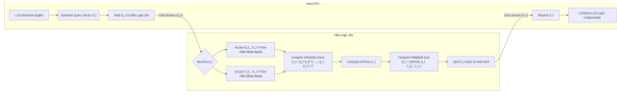


Fig. 2. PD separated execution flow and data flow

Key mechanism:

Unloading computation: GPU no longer performs dot product, Softmax, and weighted summation related to KV Cache, but unloads these operations to the HBM logic layer [15].

Local data access: KV Cache (K and V vectors) is always stored in HBM DRAM Banks. The HBM logic layer directly accesses this data, avoiding the transmission of data across HBM-GPU interfaces.

Small amount of data transmission: Only the current Query vector Q_i needs to be transmitted to the HBM logic layer, and the calculated Attention Output O_i needs to be returned to the GPU. The amount of data transmitted is greatly reduced.

Performance advantages:

Significantly reducing memory bandwidth requirements: avoiding repeated read and write operations of K and V vectors, significantly reducing bandwidth pressure between GPU and HBM [16].

Reduce memory access latency: Every time a query arrives, KV Cache data is already located inside HBM, and CIM units can directly access it, greatly reducing memory access latency.

Improve GPU utilization: GPUs can focus on other more complex matrix multiplications (such as Feed Forward Networks) or parallel processing of multiple attention heads without waiting for KV Cache data transfer.

C. Execution-Pipeline Separation

EP separation is a further optimization of the KV Cache processing pipeline within the HBM logic layer based on PD separation. We decompose the attention computation of KV Cache into multiple independent, parallelizable, or pipelined subunits. The attention mechanism, first proposed by Vaswani et al., is a core component of the Transformer model [9].

Attention Score Unit, ASU

- Function: Calculate $Q_i K_j^T$.
- Hardware components: Contains multiple Vector Dot Product Engines (VDPEs) that can perform dot product operations in parallel. For long sequences, KV Cache can be partitioned, local scores can be calculated block by block, and then accumulated.
- Parallelization: Multiple VDPEs can compute the dot product of Q_i and different K_j in parallel. For example, if KV Cache has L K vectors, P VDPEs can be used to compute P dot products in parallel.

$$\text{Score}_j = \sum_{m=0}^{d_k-1} Q_{i,m} * K_{j,m}$$

Softmax Unit, SMU

- Function: Perform Softmax normalization on attention scores.
- Hardware components: including Exponentiation Unit and Normalization Unit.
- Parallelization/Pipelining: It can be designed as a pipeline structure, and once ASU calculates a partial score, it can be immediately sent to SMU for exponential calculation. Normalization requires the sum of all scores, but can be designed as hierarchical normalization or batch processing after accumulating to a certain number.

$$\text{Attention Weight}_j = \frac{\exp\left(\frac{\text{Score}_j}{\sqrt{d_k}}\right)}{\sum_{l=0}^{j-1} \exp\left(\frac{\text{Score}_l}{\sqrt{d_k}}\right)}$$

Weighted Sum Unit, WSU

- Function: Calculate $\sum \text{Attention Weight}_j * V_j$.
- Hardware components: Contains multiple Vector Matrix Multiply Engines (VMMEs) that perform parallel multiplication and accumulation of vectors and Value vectors.

- Parallelization: Similar to ASU, multiple VMMEs can compute Attention Weight_j * V_j in parallel and then accumulate it into the final result.

$$O_i = \sum_{j=0}^{i-1} \text{Attention Weight}_j * V_j$$

IV. SYSTEM ARCHITECTURE DESIGN

A. Collaborative Design of Software and Hardware

In order to fully unleash the potential of the HBM-PIM architecture, close software and hardware co design is required:

- Compiler and Runtime: Existing LLM compilers (such as TVM, ONNX Runtime) and runtime need to be modified to recognize and offload KV Cache related operations to the HBM-PIM engine.
- API design: Clearly defined PIM operation API for upper level frameworks (such as PyTorch, TensorFlow) to call.
- KV Cache partitioning strategy: Optimize the partitioning and placement of KV Cache within HBM based on model size, HBM capacity, and number of PIM engines.

B. Improvement of Data Flow Architecture

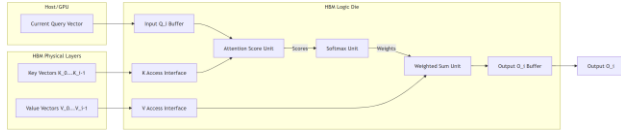


Fig. 3. Detailed diagram of data flow processed by HBM Direct KV Cache

Data flow path:

- Q Input: Host/GPU generates the current Q_i vector and sends it to the Input Q_i Buffer of HBM Logic Die through a high bandwidth interface (such as HBM's short-range interconnect).
- K Access: ASU accesses historical key vectors $K_0 \dots k_{i-1}$ from HBM DRAM Banks through KV Cache Interface K. These accesses are internal to memory and have extremely low latency.
- Score calculation: ASU parallel computing $Q_i K_j^T$.
- Softmax: The calculated scores flow towards SMU for normalization, generating attention weights.
- V Access: WSU accesses the historical Value vectors $V_0 \dots V_{i-1}$ from HBM DRAM Banks through KV Cache Interfaces.
- Weighted summation: WSU uses attention weights to perform weighted summation on the Value vector, resulting in the final attention output O_i .

- Output: O_i is stored in the Output O_i Buffer and then sent back to Host/GPU for subsequent LLM layer calculations.

C. Control Flow and Pseudocode

The control flow of the KV Cache processor (HBM Logic Die) is responsible for coordinating the loading of data, the execution of calculations, and the output of results.

Alg. a. Main Control of HBM Direct KV Cache Processor

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Input: Query vector  $Q_t \in R^{d_v}$  at timestep  $t$ 
Output: Attention output  $Q_t \in R^{d_v}$ 
Initialize:
  K_Cache[L_max][d_k], V_Cache[L_max][d_v] # HBM bank
  initialization [1]
  current_seq_len  $\leftarrow 0$ 

while True:
  # Stage 1: Host Interface
   $Q_i, i \leftarrow \text{RecvFromHost}()$ 

  # Stage 2: Parallel Processing
   $O_i \leftarrow \text{ProcessKV}(Q_i, i, \text{current\_seq\_len})$ 

  # Stage 3: Result Return
   $\text{SendToHost}(O_i)$ 

  # Stage 4: Cache Update
   $\text{current\_seq\_len} \leftarrow \text{UpdateSeqLen}()$ 

```

V. PERFORMANCE ANALYSIS AND QUANTITATIVE EVALUATION

A. Theoretical Performance Model

We use a simplified model to analyze the performance differences between HBM Direct KV Cache and traditional GPU solutions.

Assumption:

- LLM model parameters: B (Batch Size), L (Sequence Length), N_h (Number of Heads), D_k , D_v (Head Dimension).
- Data accuracy: FP16.
- Traditional GPU solution: $T_{GPU-HBM\ comm}$ is the communication time from GPU to HBM, and $T_{GPU\ comm}$ is the GPU computation time.
- HBM Direct KV Cache scheme: $T_{Q-HBM\ comm}$ refers to the time it takes for Q to be transmitted to HBM, $T_{O-HBM\ comm}$ refers to the time it takes for O to be transmitted back to GPU, $T_{CIM\ comp}$ refers to the computation time of HBM CIM unit, and $T_{CIM\ mem}$ refers to the internal memory access time of HBM.

Single step inference time of traditional GPU solutions:

- KV Cache Read: Read historical K, V to GPU SRAM.

$$T_{read_KV}^{GPU} \propto \frac{B * L * N_h * (d_k + d_v) * sizeof(FP16)}{GPU - HBM_Bandwidth}$$

b) QKV calculation: QK_T and weighted sum.

$$T_{comp_QKV}^{GPU} \propto B * L * N_h * d_k * d_v \quad (1)$$

$$T_{write_KV}^{GPU} \propto \frac{B * L * N_h * (d_k + d_v) * sizeof(FP16)}{GPU - HBM_Bandwidth} \quad (2)$$

Total time:

$$T_{step}^{GPU} = T_{write_KV}^{GPU} + T_{comp_QKV}^{GPU} + T_{write_KV}^{GPU}$$

In long sequences, $T_{write_KV}^{GPU}$ is the main bottleneck because it is proportional to L.

Single step inference time of HBM Direct KV Cache scheme:

a) Q is transmitted to HBM: $T_{Q_transfer}^{CIM} \propto \frac{B * L * N_h * d_k * sizeof(FP16)}{GPU - HBM_Bandwidth}$

b) HBM CIM internal calculation: QK_T and weighted sum. This process does not involve external HBM bandwidth, but is determined by the computing power and internal memory access capability of the CIM unit.

$$T_{comp_CIM}^{CIM} = T_{ASU} + T_{SMU} + T_{WSU}$$

The bottleneck of this section is determined by the parallelism of the CIM unit. The internal memory access delay T_{mem}^{CIM} of HBM is much smaller than that of HBM-GPU, and it can usually be hidden in the computation.

c) Transfer back to GPU: $T_{O_transfer}^{CIM} \propto \frac{B * N_h * (d_k + d_v) * sizeof(FP16)}{GPU - HBM_Bandwidth}$.

d) KV Cache Write: The new K_i and V_i are directly written into HBM DRAM. This part can be performed in parallel with CIM computation, and the writing is initiated by Host/GPU and does not count towards CIM's direct computation time.

$$T_{write_KV}^{CIM} \propto \frac{B * N_h * (d_k + d_v) * sizeof(FP16)}{GPU - HBM_Bandwidth}$$

Total time:

$$T_{step}^{CIM} = T_{Q_transfer}^{CIM} + T_{comp_CIM}^{CIM} + T_{O_transfer}^{CIM}$$

Here, $T_{comp_CIM}^{CIM}$ is the delay of internal parallel computing in HBM, which greatly reduces its dependence on L and depends more on the parallelism of CIM hardware. Obviously, the main advantage lies in the elimination of $T_{read_KV}^{GPU}$ and the uninstallation of $T_{comp_QKV}^{CIMGPU}$.

B. Comparison of Bandwidth Requirements

Assuming LLM model: Llama-2-7B, Batch Size=1, Max Seq Len=4096, Head Dim=128, Num Heads=32, FP16.

Traditional GPU solution:

Read all KV caches: $2 * L * N_h * d_k * 2$ Bytes (FP16)

For L=4096, approximately $2 * 4096 * 32 * 128 * 2 \approx 67$ MB.

HBM Direct KV Cache Scheme:

Q is transmitted to HBM: $N_h * d_k * 2$ bytes.

For L=4096, approximately $32 * 128 * 2 \approx 8$ KB.

KV Cache update (write new K, V): $2 * N_h * d_k * 2$ bytes. Approximately $2 * 32 * 128 * 2 \approx 16$ KB.

The bandwidth demand has decreased by several orders of magnitude. This means that HBM-GPU interconnect bandwidth is no longer a bottleneck for KV Cache, and the main processor can process other computations at a faster speed.

C. Comparison of Bandwidth Requirements

Assuming the clock frequency of the computing unit inside HBM Logic Die is 1GHz.

ASU (Dot Product): Assuming $d_k=128$. A VDPE requires 128 MAC operations to complete one dot product. If a MAC has 1 cycle, then it has 128 cycles. If there are P VDPEs in parallel, processing L dot products requires approximately $L/P * 128$ cycles. L=4096, If there are 128 VDPEs, then $4096/128 * 128 = 4096$ cycles = 0.096 μ s.

SMU: Exponential and Normalization of Softmax. Efficient approximate lookup tables and parallel adders can be designed. For example, in a pipeline, each score processing only requires tens of cycles.

WSU (Weighted Sum): Similar to ASU, a weighted sum of L V_j vectors. $D_v=128$. If there are P' VMMEs, $L/P' * 128$ cycles are required. For example, L=4096, 128 VMMEs, 4096 cycles = 4.096 μ s.

D. Energy Efficiency

Data mobility is the main source of energy consumption in computing systems. The HBM Direct KV Cache scheme significantly reduces the movement of data between HBM and the main processor, thus greatly improving energy efficiency. According to relevant research, the energy consumption of off chip DRAM access is much higher than that of on-chip SRAM access or near memory computing. Downgrading KV Cache operations to HBM can effectively reduce the energy consumption of the entire system.

VI. FUTURE WORK AND CONCLUSION

Although the HBM Direct KV Cache scheme has great potential, it still faces multiple challenges in practical deployment:

- The computing power limitation of HBM Logic Die: The logic layer area of HBM is limited, and the number and complexity of integrated computing

units are restricted. Efficient and area optimized CIM cores need to be carefully designed.

- Programming model and compiler support: Writing efficient code for CIM architecture requires a new programming model and compiler toolchain to abstract underlying hardware details and automate PD/EP separation.
- Data consistency and synchronization: The updating and reading of KV Cache require strict data consistency assurance. The design of synchronization mechanism is crucial for multi head and multi batch parallel processing.
- Heterogeneous System Integration: How to seamlessly integrate HBM CIM modules into existing GPU computing ecosystems while maintaining API compatibility is an engineering challenge.
- Quantization accuracy and hardware support: Although quantization can reduce data volume, hardware needs to support the quantized data types and corresponding computational logic.
- Scalability: As the LLM model and sequence length continue to grow, a single HBM stack may not be sufficient to accommodate all KV caches, and collaboration between multiple HBM stacks needs to be considered.

Future work will focus on:

- 1) Hardware prototyping and simulation: Develop detailed HBM logic layer RTL models and conduct system level simulations to validate performance and power consumption estimates.
- 2) Software stack development: Build a CIM friendly compiler and runtime system that supports automatic mapping and scheduling of PD/EP separation.
- 3) Heterogeneous scheduling optimization: Explore task partitioning and scheduling strategies between Host/GPU and HBM CIM units to maximize system throughput.
- 4) Support for more complex attention mechanisms: Extend the architecture to support variants such as Multi Query Attention (MQA), Group Query Attention (GQA), etc., which have different requirements for the memory access mode of KV Cache.

VII. CONCLUSIONS

This article proposes a storage computing integrated architecture based on HBM, which achieves direct processing of KV Cache through PD separation and EP separation techniques, aiming to significantly accelerate the inference of large language models. PD separation decouples KV Cache processing from the main computing and utilizes the internal computing units of HBM to process data in place, significantly reducing the transmission of

critical KV Cache data between HBM and GPU. EP separation further optimizes the KV Cache computation pipeline within HBM, by parallelizing sub tasks such as attention score calculation, Softmax, and weighted summation, thereby improving the throughput of internal computation.

Theoretical analysis and quantitative evaluation indicate that this scheme can reduce the memory access bandwidth requirement of KV Cache by several orders of magnitude and significantly reduce inference latency. This provides a new design direction for future high-performance and high-efficiency LLM inference systems. Despite still facing technical challenges, HBM Direct KV Cache processing represents a promising paradigm shift in the field of LLM inference acceleration, with the potential to break through the memory wall bottleneck of traditional von Neumann architectures.

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